

Establishing the Onset of Obscuration: A JWST Spectroscopic Census of High- z Dusty Star-Forming Galaxies

The total cosmic star formation rate density (SFRD) of the Universe is a critical metric of galactic growth over time. Dust obscuration hides a substantial fraction of star formation at early cosmic times, making observations of dusty star-forming galaxies (DSFGs) integral for determining the obscured contribution to the SFRD. The obscured fraction of the SFRD is key to distinguishing between competing models of a dust-rich vs dust-poor early Universe. However, the abundance of DSFGs at $z > 4$ is poorly constrained due to their relative rarity, faintness, and uncertainties in photometric analyses. As a result, the obscured SFRD during the reionization era remains highly uncertain. We propose JWST NIRSpec/MSA spectroscopic follow-up of 15 ALMA 2mm-selected DSFGs at $z = 4\text{--}7$ across 8 massive galaxy clusters to constrain the obscured star-formation history at the Epoch of Reionization with unprecedented accuracy. NIRSpec spectroscopy will secure redshifts via rest-frame optical emission lines, breaking photometric degeneracies and enabling robust characterization of the internal structure and dust content of these rare galaxies. This program will provide the largest spectroscopically confirmed sample of DSFGs at $z > 4$ to date. We will precisely measure the obscured SFRD and resolve the long-standing debate between a dust-rich vs dust-poor early Universe. This program will further establish the epoch at which obscured star formation dominates the SFRD, enabling new tests of the timeline for massive galaxy formation and growth in the early Universe.

- **Scientific Justification**

A Census of Dust-Obscured Star Formation in the Early Universe is Missing

The evolution of galaxies across cosmic time can be characterized by the star formation rate density (SFRD) of the Universe. The SFRD is composed of both unobscured star-formation (probed through UV and optical surveys) and dust obscured star-formation (traced through IR and radio surveys). While the obscured component is observed to peak at $z \sim 2$, it has been known for decades that dust continues to conceal a large fraction of star-formation at earlier epochs (e.g. Hauser et al. 1998). The obscured fraction of the SFRD is measured using the infrared luminosities of dusty star-forming galaxies (DSFGs) (Casey et al. 2014). At high redshift, the obscured fraction indicates whether the star formation in the early Universe is dominated by rare starbursts from massive galaxies or by more numerous UV bright galaxies with moderate star formation rates (Casey et al. 2018). Precisely measuring the obscured fraction of the SFRD is essential for distinguishing between competing models of a dust-rich vs. dust-poor early Universe.

However, the obscured SFRD at $z > 4$ is highly uncertain due to the poorly constrained abundance of DSFGs at this epoch (e.g. Sun et al. 2025, Zavala et al. 2021). Not only are these galaxies rare and difficult to find in blind searches, but the current known sample of DSFGs is further limited by large uncertainties in their photometric redshifts. The recent JWST discoveries of $z > 3$ massive quiescent galaxies and $z > 10$ “blue monsters” are challenging existing models of massive galaxy assembly (Ferrera et al. 2025). An understanding of star formation rates at this epoch is vital to interpret these emerging results.

We propose JWST NIRSpec/MSA spectroscopic follow-up of 15 ALMA 2 mm-selected DSFGs at $z=4-7$ across 8 massive galaxy clusters. JWST NIRSpec is the only instrument with the wavelength range and sensitivity required to efficiently obtain spectroscopic redshifts of high- z DSFGs. Redshifts detected via optical emission lines will be used alongside existing NIRCам photometry of the targets to generate an infrared luminosity function and derive the obscured SFRD. This program will deliver the most accurate constraint on the obscured SFRD at $z > 4$ to date and settle the long debate between a dust-poor vs. dust-rich early Universe. These observations will not only provide valuable test-beds for models of DSFG formation but for massive galaxy evolution as a whole across cosmic time.

Photometric Uncertainties Limit Measurements of the Obscured SFRD

The obscured fraction of the SFRD across cosmic time relies on accurate number densities of DSFGs. *However, the current sample of high- z DSFGs is limited by the relative rarity of these galaxies and the strong degeneracies between dust attenuation and photometric redshift.* For example, spectroscopic follow-up by Jin et al. 2024 proved that the DSFG candidate COSBO-7 has a redshift $z=2.63$ despite four independent SED fitting codes favoring photometric redshifts > 7 . Photometric redshift uncertainties and small sample sizes of DSFGs can result in large differences between measurements of the obscured SFRD, as demonstrated by the ALMA programs ALPINE and ASPECS whose measurements of the obscured SFRD differ by an order of magnitude (Gruppioni et al. 2020, Bouwens et al. 2020). Accurate number densities of DSFGs directly probe the limits of star formation in the early Universe.

Existing studies of the obscured SFRD and high- z IR luminosity function are missing vital spectroscopic information required to accurately trace dust obscuration through the Universe's history. We will follow-up 15 promising high- z DSFG candidates with JWST NIRSpec/MSA to obtain the largest spectroscopically secure sample of high- z DSFGs to date.

This Proposal: Spectroscopic Follow-Up of Dusty Star-Forming Galaxies in the EoR

Our targets represent the strongest high- z DSFG candidates from an ALMA 2mm survey of 60 massive lensing clusters. 2mm observations already preferentially select high- z DSFGs due to the strong negative K-correction in 2mm, in which the same galaxy appears 3x brighter at $z=6$ than $z=1$ (Casey et al. 2014). Our high- z DSFG candidates were then refined based on their 2mm/near-IR flux ratios, which should be greater for more distant, dustier galaxies (e.g. Sun et al. 2025). Flux ratios and photo- z s were computed using NIRCам observations of all 60 clusters in 10 filters from the JWST Cycle 4 VENUS program [PI: S. Fujimoto]. Using our 2mm observations and existing rich NIRCам coverage, we selected the 15 most promising high- z candidates (likely $z=4$ –7) to follow-up with NIRSpec. While ground-based interferometers like ALMA can observe DSFGs, they require prohibitively long on-source integration times to secure redshifts via molecular line scans (4–8x longer than JWST). NIRSpec is the only instrument with the wavelength range and spatial resolution required to obtain spectroscopic redshifts of our targets.

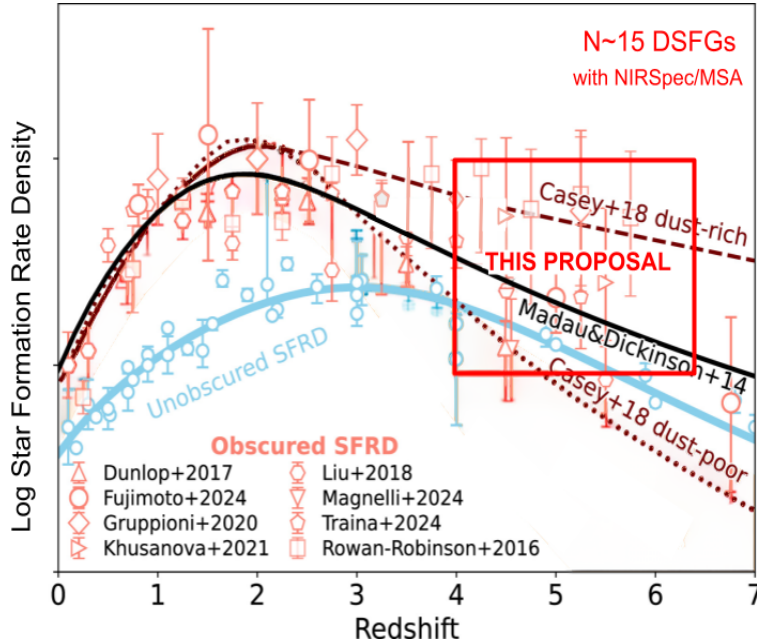


Figure 1. Spectroscopic follow-up of dusty star forming galaxies (DSFGs) in the Epoch of Reionization (EoR) is essential to constrain models of early dust production.

Existing measurements of the obscured and unobscured star-formation rate density (SFRD) are shown in open salmon symbols and light blue, respectively (Sun et al. 2025). Predictions for dust-rich and dust-poor early-Universe scenarios are indicated by the maroon lines and the best-fit total SFRD is shown by the solid black line. The parameter space probed by this program is outlined in red. Previous measurements of the obscured

SFRD at these redshifts have been comparable to the Madau & Dickinson line, suggesting that the total SFRD during the EoR may be underestimated. However, high- z measurements of the obscured SFRD are both sparse and highly uncertain, primarily due to limited spectroscopic confirmation of DSFGs. This program will obtain NIRSpec/MSA spectroscopy for ~15 newly ALMA-identified DSFGs at $z>4$, delivering the most precise constraints to date on the obscured star-formation history and resolving the long-standing debate between a dust-poor and dust-rich early Universe.

We propose NIRSpec/MSA spectroscopic follow-up of 15 candidate high- z DSFGs across 8 massive lensing clusters. We will obtain spectroscopic redshifts through detection of rest-frame optical emission lines, namely H-alpha and O[III]. Spectroscopic redshifts will break photometric degeneracies and enable accurate determinations of the number density of DSFGs as a function of redshift. This sample will double the previous largest spectroscopically secure sample of DSFGs at these redshifts (Sun et al. 2025). *As a result, this program will provide the most precise measurement of the obscured SFRD to date at $z > 4$, a parameter space that has been plagued with large uncertainties from photometric redshifts* (see Figure 1).

With a spectroscopically-secure sample of DSFGs, we will compute the IR luminosity function of our sample to derive the obscured SFRD at EoR redshifts. By reducing the large uncertainties from photometric redshifts, our measurement of the obscured SFRD will have the accuracy to differentiate between the dust-rich and dust-poor early Universe models.

DSFGs Will Improve Understanding of Early Massive Galaxy Evolution

Without understanding how much of the early Universe is dust-obscured, we cannot claim to understand the processes that shaped the local Universe. Accurate determinations of the obscured SFRD and dust content of the high- z Universe will provide a vital constraint for testing models of massive galaxy formation and evolution.

Despite substantial efforts to map the evolution of the SFRD with redshift, *recent measurements of the obscured SFRD with JWST imply that dust-obscured star formation may have been historically underestimated* (see Figure 1). The surprising observations of $z > 3$ massive, quiescent galaxies and massive $z > 10$ “blue monster” galaxies with JWST also hint that star formation may have proceeded more rapidly than previously suggested (Ferrera et al. 2025). The obscured SFRD is directly linked to how these galaxies emerged and addresses the JWST’s science goal of understanding “how did the first galaxies form?” DSFGs have been proposed as an evolutionary link between these massive quiescent and blue monster galaxies. Not only will our observations enable robust physical characterization of the most rapidly star forming galaxies at the EoR, but it will enable further studies on the possible link between JWST’s most surprising high- z discoveries. Finally, this work will distinguish between the dust-poor vs dust-rich early Universe models, facilitating accurate future modeling of dust mass assembly, reionization, and galaxy evolution across cosmic time.

References

- Hauser, M. G. et al., *ApJ* 508, 1 (1998).
- Casey, C. M. et al., *Physics Reports* 541, 2 (2014).
- Casey, C. M. et al., *ApJ* 862, 1 (2018).
- Sun, F. et al., *ApJ* 980, 1 (2025).
- Zavala, J. A. et al., *ApJ* 909, 2 (2021).
- Ferrera, A. et al., *A&A* 694, 286 (2025).
- Jin, S. et al., *A&A* 960, 16 (2024).
- Gruppioni, C. et al., *A&A* 643, 8 (2020).
- Bouwens, R. et al., *ApJ* 902, 2 (2020).